

$T = 1,000$ K (to measure the robustness of the learned potentials). We trained two types of model: (1) a NequIP model from scratch and (2) a fine-tuned model that was pretrained on the GNoME dataset, starting from the checkpoint before the learning rate was reduced the first time. The network architecture is identical to that used in pretraining. Because the AIMD data contain fewer high-force/high-energy configurations, we use a L2 loss in the joint loss function instead of a Huber loss, again with $\lambda_E = 1.0$ and $\lambda_F = 0.05$. For all training set sizes and all materials, we scan learning rates 1×10^{-2} and 2×10^{-3} and batch sizes 1 and 16. Models are trained for a maximum of 1,000 epochs. The learning rate is reduced by a factor of 0.8 if the test error on a hold-out set did not improve for 50 epochs. We choose the best of these hyperparameters based on the performance of the final checkpoint on the 400-K test set. The 400-K test set is created using the final part of the AIMD trajectory. The training sets are created by sampling varying training set sizes from the initial part of the AIMD trajectory. The out-of-distribution robustness test is generated from the AIMD trajectory at 1,000 K. Training is performed on a single V100 GPU.

Molecular dynamics simulations. The materials for AIMD simulation are chosen on the basis of the following criteria: we select all materials in the GNoME database that are stable, contain one of the conducting species under consideration (Li, Mg, Ca, K, Na) and have a computationally predicted band gap >1 eV. The last criterion is chosen to not include materials with notable electronic conductivity, a desirable criterion in the search for electrolytes. Materials are run in their pristine structure, that is, without vacancies or stuffing. The AIMD simulations were performed using the VASP. The temperature is initialized at $T = 300$ K, ramped up over a time span of 5 ps to the target temperature, using velocity rescaling. This is followed by a 45-ps simulation equilibration using a Nosé–Hoover thermostat in the NVT ensemble. Simulations are performed at a 2-fs time step.

Machine-learning-driven molecular dynamics simulations using JAX MD⁶³ are run on a subset of materials for which AIMD data were available and for which the composition was in the test set of the pretraining data (that is, previously unseen compositions), containing Li, Na, K, Mg and Ca as potentially conducting species. This results in 623 materials for which GNoME-driven molecular dynamics simulations are run. Simulations are performed at $T = 1,000$ K using a Nosé–Hoover thermostat, a temperature equilibration constant of 40 time steps, a 2-fs time step and a total simulation length of 50 ps. Molecular dynamics simulations are performed on a single P100 GPU.

For analysis of both the AIMD and the machine learning molecular dynamics simulation, the first 10 ps of the simulation are discarded for equilibration. From the final 40 ps, we compute the diffusivity using the DiffusionAnalyzer class of pymatgen with the default smoothed=max setting^{23,70,71}.

Data availability

Crystal structures corresponding to stable discoveries discussed throughout the paper will be made available at https://github.com/google-deepmind/materials_discovery. In particular, we provide results for all stable structures, as well as any material that has been recomputed from previous datasets to ensure consistent settings. Associated data from the r^2 SCAN functional will be provided, expectantly serving as a foundation for analysing discrepancies between functional choices. Data will also be available via the Materials Project

at <https://materialsproject.org/gnome> with permanent link: <https://doi.org/10.17188/2009989>.

Code availability

Software to analyse stable crystals and associated phase diagrams, as well as the software implementation of the static GNN and the interatomic potentials, will be made available at https://github.com/google-deepmind/materials_discovery.

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Competing interests Google LLC owns intellectual property rights related to this work, including, potentially, patent rights.

Additional information

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